

$$3d^8$$

September 1, 2026

$$1 \quad \left| 3d_{xz}^2 3d_{yz}^2 3d_{x^2-y^2}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha \right\rangle$$

Here are the CSFs:

$$|1, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|2, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|3, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|4, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|5, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|6, 1, 1\rangle = \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle$$

$$|7, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|8, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle$$

$$|9, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle$$

$$|10, 1, 1\rangle = \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle$$

$$|11, 1, 0\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle$$

$$|12, 1, 0\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle$$

$$\begin{aligned}
|13, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|14, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|15, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|16, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|17, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|18, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|19, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|20, 1, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|21, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|22, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|23, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|24, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|25, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|26, 1, -1\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|27, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|28, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|29, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|30, 1, -1\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|31, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|32, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|33, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|34, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|35, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|36, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|37, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|38, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|39, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|40, 0, 0\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|41, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|42, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|43, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|44, 0, 0\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|45, 0, 0\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 3 13 23

$$\begin{aligned}
\langle 3, 1, 1 | L_x | 6, 1, 1 \rangle &= -\sqrt{3}I \\
\langle 3, 1, 1 | L_x | 8, 1, 1 \rangle &= -I
\end{aligned}$$

$$\Delta g_{xx}$$

$$\begin{aligned}
&+3\zeta\Delta_{3-6}^{-1} \\
&+\zeta\Delta_{3-8}^{-1}
\end{aligned}$$

$$\begin{aligned}\langle 3, 1, 1 | L_y | 5, 1, 1 \rangle &= I \\ \langle 3, 1, 1 | L_y | 9, 1, 1 \rangle &= \sqrt{3}I\end{aligned}$$

$$\Delta g_{yy}$$

$$\begin{aligned}&+ \zeta \Delta_{3-5}^{-1} \\ &+ 3\zeta \Delta_{3-9}^{-1}\end{aligned}$$

$$\langle 3, 1, 1 | L_z | 1, 1, 1 \rangle = -2I$$

$$\Delta g_{zz}$$

$$+ 4\zeta \Delta_{3-1}^{-1}$$

For a multiplicity of 3 we have the following:

$$D = \langle 1, 1 | 1, 1 \rangle - \langle 1, 0 | 1, 0 \rangle$$

$$E = \langle 1, -1 | 1, 1 \rangle$$

$$\begin{aligned}\langle 1, 1 | \hat{H}_{eff} | 1, 1 \rangle &= -\zeta^2 (\\ &+ \Delta_1^{-1} + \frac{1}{8}\Delta_{15}^{-1} + \frac{3}{8}\Delta_{16}^{-1} + \frac{1}{8}\Delta_{18}^{-1} + \frac{3}{8}\Delta_{19}^{-1} + \frac{1}{8}\Delta_{35}^{-1} \\ &+ \frac{3}{8}\Delta_{36}^{-1} + \frac{1}{8}\Delta_{38}^{-1} + \frac{3}{8}\Delta_{39}^{-1})\end{aligned}$$

$$\begin{aligned}\langle 1, 0 | \hat{H}_{eff} | 1, 0 \rangle &= -\zeta^2 (\\ &+ \frac{1}{8}\Delta_5^{-1} + \frac{3}{8}\Delta_6^{-1} + \frac{1}{8}\Delta_8^{-1} + \frac{3}{8}\Delta_9^{-1} + \frac{1}{8}\Delta_{25}^{-1} + \frac{3}{8}\Delta_{26}^{-1} \\ &+ \frac{1}{8}\Delta_{28}^{-1} + \frac{3}{8}\Delta_{29}^{-1} + \Delta_{31}^{-1})\end{aligned}$$

$$\begin{aligned}\langle 1, -1 | \hat{H}_{eff} | 1, 1 \rangle &= -\zeta^2 (\\ &- \frac{1}{8}\Delta_{15}^{-1} + \frac{3}{8}\Delta_{16}^{-1} + \frac{1}{8}\Delta_{18}^{-1} - \frac{3}{8}\Delta_{19}^{-1} + \frac{1}{8}\Delta_{35}^{-1} \\ &- \frac{3}{8}\Delta_{36}^{-1} - \frac{1}{8}\Delta_{38}^{-1} + \frac{3}{8}\Delta_{39}^{-1})\end{aligned}$$

$$\begin{aligned}
D = & \zeta^2 (\\
& -\Delta_1^{-1} + \frac{1}{8}\Delta_5^{-1} + \frac{3}{8}\Delta_6^{-1} + \frac{1}{8}\Delta_8^{-1} + \frac{3}{8}\Delta_9^{-1} - \frac{1}{8}\Delta_{15}^{-1} \\
& - \frac{3}{8}\Delta_{16}^{-1} - \frac{1}{8}\Delta_{18}^{-1} - \frac{3}{8}\Delta_{19}^{-1} + \frac{1}{8}\Delta_{25}^{-1} + \frac{3}{8}\Delta_{26}^{-1} + \frac{1}{8}\Delta_{28}^{-1} \\
& + \frac{3}{8}\Delta_{29}^{-1} + \Delta_{31}^{-1} - \frac{1}{8}\Delta_{35}^{-1} - \frac{3}{8}\Delta_{36}^{-1} - \frac{1}{8}\Delta_{38}^{-1} - \frac{3}{8}\Delta_{39}^{-1} \\
&)
\end{aligned}$$

$$\begin{aligned}
E = & \zeta^2 (\\
& + \frac{1}{8}\Delta_{15}^{-1} - \frac{3}{8}\Delta_{16}^{-1} - \frac{1}{8}\Delta_{18}^{-1} + \frac{3}{8}\Delta_{19}^{-1} - \frac{1}{8}\Delta_{35}^{-1} \\
& + \frac{3}{8}\Delta_{36}^{-1} + \frac{1}{8}\Delta_{38}^{-1} - \frac{3}{8}\Delta_{39}^{-1})
\end{aligned}$$